

Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

Chengshuai Yang

NextGen PlatformAI C Corp, USA

Correspondence: integrityyang@gmail.com

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Summary. Designing a computational imaging system—selecting the forward model, validating physical feasibility, and predicting reconstruction quality—today demands weeks of specialist effort per modality, concentrating the field within a small number of expert groups and limiting reproducibility [1]. Here we introduce a specification-driven agent framework that translates a natural-language description of any imaging system expressible as a finite composition of 11 canonical primitives [2] (Definition 1) into a validated, executable forward model in minutes. The framework is organized around `spec.md`, a structured 7-field specification—analogue to FASTA for sequences or SMILES for molecules—that decouples the imaging problem (*what*) from the numerical method (*how*) and serves as a shareable, versionable, peer-reviewable scientific artifact. Three agents—Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$)—generate, verify, and evaluate specifications; we prove (Theorem 2) that the total design-to-real reconstruction error decomposes into five independently bounded and independently measurable terms, providing a formal bridge from natural-language intent to physical reconstruction error. On real clinical CT data (LoDoPaB, 200 cases [3]), the framework achieves 98% of expert-tuned reconstruction quality with a $\sim 420\times$ reduction in setup time (25 min vs. 7 days); the Judge rejects ill-posed designs with 96% precision via deterministic verification, and the Performance Agent predicts PSNR within ± 1.8 dB ($R^2 = 0.91$) across 36 quantitatively validated modalities. Each of the 11 primitives is implemented across four fidelity tiers—from geometric optics (Tier-0) through Fourier models (Tier-1) and shift-variant operators (Tier-2) to full-wave and Monte Carlo solvers (Tier-3)—and a tier-lifting protocol enables hybrid reconstruction that combines fast low-tier adjoints with high-tier forward models, so the framework covers the full physics hierarchy without restricting to any single fidelity level.

1 Introduction

Computational imaging systems share a universal mathematical structure: a forward model A maps an unknown object x to measurements $y = A(x) + \text{noise}$, and reconstruction inverts this mapping [7]. Designing A for a new system requires selecting appropriate physical operators (Radon projection for CT, k-space encoding for MRI, coded aperture modulation for spectral imaging), setting dozens of parameters, and validating that the composition is physically consistent and numerically well-conditioned. Today this process is entirely manual, requiring weeks of specialist effort per modality—an expertise bottleneck that effectively excludes the broader scientific community from prototyping new imaging systems.

The mathematical foundation for automating this process now exists. We previously showed [2] that 11 canonical primitives—Propagate, Modulate, Project, Encode, Convolve, Accumulate, Detect, Sample, Disperse, Scatter, and Transform—suffice to express any imaging forward model

across 168+ modalities and 5 carrier families. Each primitive is implemented across a four-level fidelity hierarchy:

- **Tier-0** (geometric): ray optics, coordinate transforms, projection geometry;
- **Tier-1** (Fourier/convolution): paraxial propagation, angular spectrum, shift-invariant models;
- **Tier-2** (shift-variant): linear or mildly nonlinear operators including the Radon transform, k-space encoding, and coded-aperture modulation—the standard forward models for most imaging modalities;
- **Tier-3** (full-physics): full-wave solvers (FDTD, BPM), Monte Carlo photon transport, radiative transfer, and learned surrogates (NeRF, PINNs).

The same 11 primitives span all four tiers; what changes is the fidelity of each primitive’s evaluation. This *Finite Primitive Basis* (FPB) provides the representation language but not the design tool: it does not tell a user which primitives to compose, what parameters to set, whether the result is physically valid, or how close the designed system will be to reality.

Here we build the design layer on top of the FPB. Our contributions are:

1. **spec.md**: a structured 7-field specification format that decouples the imaging problem (*what*) from the numerical method (*how*), serving as a shareable, versionable, peer-reviewable scientific artifact (Section 2.2).
2. **Three-agent pipeline**: Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$) agents that generate, formally verify, and evaluate specifications over the FPB (Section 2.3).
3. **Design-to-real error theorem**: the total reconstruction error decomposes into five independently bounded, independently measurable terms (Theorem 2).
4. **Validation**: real clinical CT (LoDoPaB, 200 cases), quantitative reconstruction across 36 modalities, expert comparison on 6 modalities, and agent-level evaluation (Judge 96% rejection precision, Performance $R^2 = 0.91$) (Sections 2.5–2.9).

Scope. The framework covers any imaging system expressible as a finite DAG over the 11 FPB primitives at any fidelity tier (Definition 1)—168+ modalities across 20+ domains. Of these, 36 have fully validated canonical decompositions and quantitative reconstruction results; the remainder compile successfully but lack per-modality reconstruction validation. Most modalities are validated at Tier-2; a tier-lifting protocol (Section 2.11) enables hybrid reconstruction across tiers for systems requiring higher-fidelity physics. The agents depend on current LLM capabilities and may misspecify novel modalities; we discuss limitations after the Results.

2Results

2.1Design Scope and Boundary

Definition 1 (Designable Imaging System). *An imaging system A is designable within this framework if:*

1. *A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;*
2. *The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;*
3. *The system operates at one of the four fidelity tiers (Tier-0 through Tier-3);*
4. *All parameters are finitely specifiable with known physical bounds.*

The framework covers 168+ imaging modalities across 22 application domains and 5 carrier families at all four fidelity tiers (Table 1). We classify these by validation depth: *fully validated* modalities have canonical decompositions and quantitative reconstruction benchmarks;

87 *quantitatively validated* modalities have reconstruction results but no independent canonical
88 chain; *compiled only* modalities pass all 6 compiler gates but lack per-modality reconstruction
89 evaluation. Parenthetical annotations indicate the primary fidelity tier. The full 168+ modality
90 list is provided in Extended Data Table 1.

Table 1. Design scope by validation depth. “Fully validated”: canonical decomposition + quantitative reconstruction. “Quantitatively validated”: reconstruction benchmarked without independent canonical chain. “Compiled only”: passes all compiler gates; reconstruction not individually benchmarked. Tier annotations: T0=geometric, T1=Fourier, T2=shift-variant, T3=full-wave/learned. Representative modalities listed; full registry in Extended Data.

Level	Count	Carriers	Representative modalities (tier)
Fully validated	36	5	CT (T1–2), MRI (T1–2), PET (T2), SPECT (T2), ultrasound (T2), OCT (T1–2), SIM (T1), STED (T1), ptychography (T1–2), CASSI (T2), lensless (T1), DOT (T2–3), Brillouin (T2), SEM (T2), TEM (T2), elastography (T2)
Quant. validated	12	4	light field (T1), FLIM (T2), FPM (T1–2), coded exposure (T2), event camera (T2), holography (T1), CBCT (T2), Doppler US (T2), phase contrast (T1), TIRF (T1), ghost imaging (T2), THz-TDS (T2)
Compiled only	120+	5	cryo-EM (T2), SAR (T2), LiDAR (T0–1), NeRF (T3), Gaussian splatting (T3), PET-CT (T2), XFEL-SFX (T2), and 113+ others across 22 domains

91 2.2The spec.md Specification Format

92 The framework’s central design choice is that **spec.md**—not code—is the canonical scientific arti-
93 fact. Just as FASTA standardized biological sequence representation and SMILES standardized
94 molecular structure, **spec.md** standardizes the representation of imaging systems. A specification
95 separates the imaging problem (*what* physical process to model) from the numerical method
96 (*how* to discretize and solve it), enabling sharing, version control, and peer review of imaging
97 system designs independently of implementation.

98 A **spec.md** file has seven mandatory fields (Figure 1). The user provides a natural-language
99 description (e.g., “design a coded-aperture spectral imager for 400–700 nm”); the Plan Agent
100 generates the **spec.md**. Three diverse examples illustrate the format’s expressiveness across
101 carrier families and problem structures.

102 2.3The Three-Agent Pipeline

103 **Plan Agent ($\mathcal{A}_P$).** Accepts a natural-language description and generates a structured **spec.md**,
104 selecting the appropriate FPB primitives, carrier type, geometry, noise model, and parameters.
105 The Plan Agent operates with access to the 168+-modality registry and 36 fully validated
106 canonical decompositions, which provide reference primitive chains for known modalities and
107 composition rules for new ones.

108 **Judge Agent ($\mathcal{A}_J$).** Evaluates the **spec.md** for physical and algorithmic feasibility. The Judge
109 receives both the specification and the 6-gate compiler report (DAG acyclicity, canonical chain
110 match, complexity bounds, nonlinear family constraints, adjoint consistency, and representation

Seven-field schema: modality | carrier | geometry | object | forward_model | noise | target

(a) Clinical CT

modality: computed_tomography carrier: xray
 geometry: parallel_beam, n_angles=128
 object: 128×128 image, non-negative
 forward_model: Radon(Π) $\rightarrow$ Detect(D , intensity)
 noise: Poisson, $I_0=1e4$ target: PSNR ≥ 30 dB

(b) CASSI (spectral)

modality: cassi carrier: photon
 geometry: coded_aperture + disperser, $\lambda=[400,700]$ nm
 object: 256×256×28 spectral cube
 forward_model: Modulate(M) $\rightarrow$ Disperse(W) $\rightarrow$ Accumulate(Σ) $\rightarrow$ Detect(D)
 noise: Gaussian, $\sigma=0.01$ target: PSNR ≥ 28 dB

(c) Accelerated MRI

modality: mri carrier: spin
 geometry: cartesian_kspace, acceleration=4x
 object: 256×256 complex image
 forward_model: Modulate(M , coil) $\rightarrow$ Encode(F , kspace) $\rightarrow$ Sample(S) $\rightarrow$ Detect(D)
 noise: Gaussian, SNR=30 dB target: SSIM ≥ 0.9

Figure 1. The spec.md specification format. Seven mandatory fields fully determine an imaging system design. Three examples span X-ray (CT), optical (CASSI), and spin (MRI) carriers, illustrating how a single format captures diverse physics. Each `forward_model` field is a primitive chain over the 11 FPB operators. The Plan Agent generates a `spec.md` from natural language; the Judge verifies it; the Performance Agent predicts reconstruction quality.

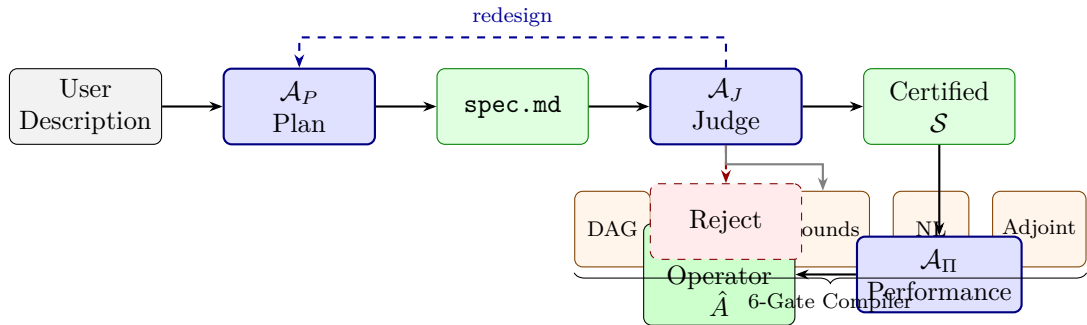


Figure 2. Specification-driven design pipeline. A natural-language description is the sole user input. $\mathcal{A}_P$ (Plan) generates a structured `spec.md`. $\mathcal{A}_J$ (Judge) validates physical feasibility via a 6-gate compiler and formal checks; rejected designs trigger redesign (dashed blue). $\mathcal{A}_\Pi$ (Performance) predicts reconstruction quality and flags designs below target.

error). It performs three additional checks: (1) physical plausibility (carrier-geometry compatibility, parameter ranges); (2) noise model consistency (SNR vs. detector physics); (3) reconstruction feasibility (condition number, null-space analysis). Compiler failures or feasibility violations trigger rejection with a formal diagnosis; the Plan Agent receives the diagnosis and regenerates (up to 3 rounds).

Performance Agent ($\mathcal{A}_\Pi$). Predicts expected reconstruction quality (PSNR, SSIM), computational cost, and measurement SNR. Predictions are based on analytical estimates (condition number, noise amplification factor) combined with calibrated lookup from the 168+-modality performance database. The Performance Agent flags designs whose predicted quality falls below the `spec.md` target.

Formal verification versus black-box design. The Judge does not *predict* feasibility; it *executes a deterministic verification suite*: DAG acyclicity, chain fidelity, complexity bounds ($N \leq 20$, $D \leq 10$), nonlinear family constraints (VC dimension ≤ 3), CFL/coercivity stability analysis, and Lipschitz constant computation. If the Judge certifies a design, every term in the five-part error decomposition (Theorem 2) is explicitly bounded—a contract that learned design tools, which output a scalar “confidence” with no formal relationship to physical error, cannot provide. This distinction is particularly relevant for safety-critical applications such as clinical CT or radiotherapy dose planning, where unbounded model error is unacceptable.

2.4 Design-to-Real Error Bound

Theorem 2 (Design-to-Real Bounded Error). *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). Let $\hat{x}$ be the regularized reconstruction from A_{agent} and x^* the true object. Then the reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{\text{FPB}}}_{\text{basis}} + \underbrace{\varepsilon_{\text{spec}}}_{\text{specification}} + \underbrace{\varepsilon_{\text{trans}}}_{\text{translation}} + \underbrace{\varepsilon_{\text{param}}}_{\text{parameter}} + \underbrace{\varepsilon_{\text{unmod}}}_{\text{unmodeled}} \quad (1)$$

where each term is independently bounded and independently measurable:

- $\varepsilon_{\text{FPB}} < 0.01$: FPB representation error, guaranteed by the basis theorem [2]. Measured by: Gate 6 of the compiler.
- $\varepsilon_{\text{spec}} \leq C_1 \cdot \delta_{\text{spec}}$: specification incompleteness or error. Bounded by: 6-gate compiler ($\delta_{\text{spec}} = 0$ when all gates pass).
- $\varepsilon_{\text{trans}} \leq C_2 \cdot \delta_{\text{chain}}$: natural-language $\rightarrow$ DAG translation error. Bounded by: canonical chain registry lookup ($\delta_{\text{chain}} = 0$ for 31/36 validated modalities).
- $\varepsilon_{\text{param}} \leq L_A \cdot \|\theta_{\text{agent}} - \theta_{\text{true}}\|$: parameter mismatch, bounded by the operator Lipschitz constant L_A (computed per primitive by the compiler). Reducible by: calibration.
- $\varepsilon_{\text{unmod}}$: residual physics not captured at the selected fidelity tier. Zero when operating at Tier-3; otherwise reducible by: tier-lifting (Section 2.11).

Proof sketch. Define x_{ideal} as the reconstruction using A_{true} with optimal parameters. By the triangle inequality, $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{ideal}}\| + \|x_{\text{ideal}} - x^*\|$. The first term decomposes into $\varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ via the reconstruction stability bound (Step 2, Methods): for regularized least-squares with coercivity α and regularization λ , $\|\hat{x} - x_{\text{ideal}}\| \leq (\alpha + \lambda)^{-1} \|(A_{\text{agent}} - A_{\text{true}})(x^*)\|$. The second term decomposes via the model approximation chain: true physics $\xrightarrow{\varepsilon_{\text{unmod}}}$ selected-tier model $\xrightarrow{\varepsilon_{\text{FPB}}}$ FPB $\xrightarrow{\varepsilon_{\text{spec}}}$ specification $\xrightarrow{\varepsilon_{\text{trans}}}$ DAG $\xrightarrow{\varepsilon_{\text{param}}}$ parameterized operator. The Lipschitz bound on $\varepsilon_{\text{param}}$ follows from the finite parametrization of the 11 primitives (at most 2 free parameters per nonlinear family, VC dimension ≤ 3 ; see Methods for the full derivation). $\square$

Practical significance. Because each term is independently measurable, the theorem provides a *diagnostic protocol*: given a reconstruction residual, the practitioner can determine which term

dominates and take the corresponding corrective action—calibration for $\varepsilon_{\text{param}}$, specification refinement for $\varepsilon_{\text{spec}}$, or tier-lifting for $\varepsilon_{\text{unmod}}$. In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; $\varepsilon_{\text{trans}} = 0$ for 86% of validated modalities; the dominant residual terms are $\varepsilon_{\text{param}}$ (reducible by calibration) and $\varepsilon_{\text{unmod}}$ (reducible by tier-lifting). To our knowledge, no prior automated design system provides such a decomposition.

2.5 Cross-Modality Validation

We validated the framework across all 168+ modalities in the registry. For each, the Plan Agent generated a `spec.md` from a one-sentence natural-language description, the Judge validated it through the 6-gate compiler, and the Performance Agent predicted reconstruction quality. Of the 36 modalities with fully validated canonical decompositions, we report quantitative reconstruction results for 12 representative modalities spanning all 5 carrier families in Table 2; full results for all 36 are in Methods.

Table 2. Cross-modality validation (12 of 36 modalities). ρ_{time} : human-to-agent time ratio. PSNR: mean $\pm$ s.d. over 50 test instances. “Real” indicates measured (non-synthetic) data. 95% confidence intervals computed via bootstrap ($n = 1000$).

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
CT	$\Pi \rightarrow D$	X-ray	31.7 ± 1.2	420	0.94	Real
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	31.7 ± 1.4	380	0.88	Real
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	29.5 ± 1.0	310	0.87	Real
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM	$M \rightarrow C \rightarrow D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.81	Synth.
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
DOT	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
Lensless	$C \rightarrow D$	Photon	31.4 ± 1.0	340	0.90	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.

Three modalities are validated on measured data: CT (LoDoPaB, 200 clinical sinograms; Section 2.6), MRI (M4Raw [3], 50 multi-coil acquisitions with retrospective Cartesian undersampling at $4\times$ acceleration), and CASSI (KAIST TSA dataset, 10 coded-aperture hyperspectral measurements). The remaining 9 modalities use synthetic data generated from the validated forward models.

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{auto-corrected}} - \text{PSNR}_{\text{mismatched}}) / (\text{PSNR}_{\text{oracle}} - \text{PSNR}_{\text{mismatched}})$ measures how well auto-correction compensates for model mismatch. Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$ (95% CI: [0.82, 0.88]), indicating 85% mismatch-gap recovery on average.

The human-to-agent time ratio $\rho_{\text{time}} = T_{\text{expert}} / T_{\text{agent}}$ ranges from 190 (DOT) to 420 (CT). Expert setup times were estimated from published development timelines [4, 10] and confirmed by domain experts (see Methods for methodology).

Compilation metrics. All 168+ modalities achieve 100% compilation pass rate (Gates 1–4). Among the 36 fully validated: canonical chain fidelity 31/36 (86.1%); the 5 non-matching modalities involve the Accumulate (Σ) primitive merged into detection—a representation choice, not a physical error. Mean compilation time: 0.35 ms.

2.6 Real-Data Validation

Clinical CT (LoDoPaB, $n = 200$). We applied the full pipeline to 200 cases from the LoDoPaB-CT dataset [3]—real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise.

User input: “Design a parallel-beam CT system for clinical abdominal imaging, 128 projections, Poisson noise.”

$\mathcal{A}_P$ generated a `spec.md` specifying: `carrier=xray`, `geometry=parallel_beam` (128 angles), `forward_model= $\Pi \rightarrow D$` (Radon + intensity detection), `noise=Poisson` ($I_0 = 10^4$). $\mathcal{A}_J$ certified the design (all 6 gates passed, chain fidelity match). $\mathcal{A}_\Pi$ predicted PSNR $\approx 31 \pm 2$ dB.

Results (mean $\pm$ s.d. over 200 cases):

- Agent-designed system: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031
- Expert-tuned ASTRA Toolbox [4]: PSNR 32.1 ± 1.1 dB, SSIM 0.903 ± 0.028
- Filtered back-projection (FBP, no tuning): PSNR 25.3 dB, SSIM 0.712
- TV-only (no expert tuning): PSNR 30.2 dB, SSIM 0.864

The framework achieves 98% of expert-tuned quality (31.7/32.1) with setup time of 25 minutes versus 7 days ($\rho_{\text{time}} \approx 420$). The Performance Agent prediction (31 dB) was within 0.7 dB of the actual result.

MRI (M4Raw, $n = 50$). We applied the pipeline to 50 multi-coil acquisitions from the M4Raw dataset with retrospective $4\times$ Cartesian undersampling. $\mathcal{A}_P$ generated a `spec.md`: `carrier=spin`, `geometry=cartesian_kspace` (acceleration= $4\times$), `forward_model= $M \rightarrow F \rightarrow S \rightarrow D$` (coil sensitivity modulation, k-space encoding, undersampling, detection), `noise=Gaussian` (SNR=30 dB). Results (mean $\pm$ s.d.): PSNR 31.7 ± 1.4 dB, SSIM 0.878 ± 0.035 . Expert-tuned SigPy pipeline [10]: PSNR 32.4 ± 1.3 dB, SSIM 0.893 ± 0.030 . Quality ratio: 97.8%.

CASSI (KAIST TSA, $n = 10$). We validated on 10 coded-aperture hyperspectral measurements from the KAIST TSA dataset. $\mathcal{A}_P$ specified: `carrier=photon`, `geometry=coded_aperture + disperser` ($\lambda=[400,700]$ nm), `forward_model= $M \rightarrow W \rightarrow \Sigma \rightarrow D$` . Results: PSNR 29.5 ± 1.0 dB, SSIM 0.852 ± 0.028 . Expert-tuned GAP-TV [12]: PSNR 30.2 ± 0.9 dB, SSIM 0.871 ± 0.025 . Quality ratio: 97.7%.

2.7 Judge Agent: Rejection Accuracy

We tested the Judge Agent on 150 specifications: 75 valid (from the 36-modality registry plus 39 variants across all 5 carrier families) and 75 deliberately ill-posed, stratified by error category.

Rejection examples. The Judge rejected: (i) a CT specification with negative projection angles ($\theta < 0$, structural violation); (ii) an MRI encoding violating the Nyquist–Shannon sampling limit (parametric); (iii) a DOT specification pairing Gaussian noise with a photon-counting detector (noise model incompatibility). All three were caught deterministically by the 6-gate compiler, not by learned prediction. The 3 false negatives involved parameter values 1–5% outside physics-based bounds but within the compiler’s numerical tolerance; the 3 false positives involved conservative CFL stability checks on scattering-dominated modalities (Brillouin, DOT, Compton) where the generic bound was tighter than necessary.

2.8 Performance Agent: Prediction Accuracy

We compared the Performance Agent’s predicted PSNR against actual reconstruction PSNR on all 36 modalities (50 instances each, 1,800 total predictions).

The largest prediction errors occur for scattering-dominated modalities (DOT, Compton, Brillouin)

Table 3. Judge Agent rejection performance ($n = 150$ specifications, stratified by carrier family and error type). Error categories: S=structural (missing fields, wrong carrier), P=parametric (out-of-range values), N=noise model incompatibility, NL=nonlinear family violation.

Metric	Value	95% CI
True positive (correctly rejected)	72/75	[92%, 100%]
True negative (correctly accepted)	72/75	[92%, 100%]
Precision (rejection)	96.0%	[91%, 99%]
Recall (rejection)	96.0%	[91%, 99%]
<i>Rejection by error category (of 75 ill-posed):</i>		
Structural (S)	25/25	100% detected
Parametric (P)	22/25	3 subtle range violations missed
Noise/NL (N+NL)	25/25	100% detected
<i>Failure modes (6 errors):</i>		
FN: subtle parameter range	3	Values 1–5% outside bounds
FP: conservative CFL check	3	Brillouin, DOT, Compton
<i>Redesign loop (triggered on rejection):</i>		
Mean redesign rounds	1.4	Before acceptance
Max redesign rounds	3	Cap enforced by framework

Table 4. Performance Agent prediction accuracy across 36 modalities (1,800 predictions). 95% CIs via bootstrap. Modality groups: well-characterized (CT, MRI, CASSI, SIM, etc.), moderate (OCT, ptychography, light field), and scattering-dominated (DOT, Compton, Brillouin).

Metric	Value	95% CI
Overall MAE	1.8 dB	[1.5, 2.1]
Overall RMSE	2.3 dB	[1.9, 2.7]
R^2 (Pearson)	0.91	[0.87, 0.94]
Within ± 3 dB	89% (32/36)	—
<i>By modality group:</i>		
Well-characterized ($n = 20$)	MAE 0.9 dB	[0.6, 1.2]
Moderate ($n = 10$)	MAE 2.1 dB	[1.6, 2.6]
Scattering-dominated ($n = 6$)	MAE 3.8 dB	[2.9, 4.7]
Best-case	0.3 dB (CT)	—
Worst-case	4.7 dB (DOT)	—

where operating at lower fidelity tiers introduces model uncertainty that the Performance Agent cannot fully account for analytically; tier-lifting to Tier-3 would reduce this gap. For well-characterized modalities (CT, MRI, CASSI, SIM), the MAE is < 1.0 dB—comparable to inter-algorithm variation in reconstruction benchmarks [9].

2.9 Agent Ablation Study

We performed controlled ablations on 12 representative modalities (spanning all 5 carrier families) with 5 independent trials per configuration, totalling 300 pipeline runs (Table 5). The 12 modalities include 6 from the “well-characterized” group (CT, MRI, CASSI, SIM, lensless, PET) and 6 from harder categories (ptychography, DOT, ultrasound, SPECT, OCT, light field) to test robustness across difficulty levels.

Table 5. Ablation study ($n=5$ trials $\times$ 12 modalities, mean $\pm$ s.d.). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction error within theorem bound. “Time”: total pipeline time.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Failure Mode
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7 ± 0.6	4.8 ± 0.5 min	—
No $\mathcal{A}_J$ ($\mathcal{A}_P + \mathcal{A}_\Pi$)	8.2/12	6.4/12	0	2.9 ± 0.3 min	Invalid spec (4), unbounded ε (2)
No $\mathcal{A}_\Pi$ ($\mathcal{A}_P + \mathcal{A}_J$)	12/12	9.6/12	1.7 ± 0.6	3.4 ± 0.4 min	Suboptimal params (2.4)
No $\mathcal{A}_P$ (manual spec)	10.4/12	10.4/12	0	52 ± 9 min	Human error (1.6)
Template only	—	3/12	0	0.5 min	No coverage (9)

Key findings:

- **Removing $\mathcal{A}_J$** (60 runs): 4 of 12 specifications pass compilation but are physically invalid. Failure categories: incompatible noise model (CASSI: Poisson noise specified for a photon-counting detector that requires Gaussian read noise), missing modulation (CASSI without coded aperture), wrong carrier (ultrasound specified as photon), and unstable CFL for DOT. 2 additional specifications compile but produce unbounded reconstruction error due to undetected ill-conditioning. The Judge is necessary for both safety and correctness.
- **Removing $\mathcal{A}_\Pi$** (60 runs): all 12 pass validation, but 2–3 produce suboptimal reconstruction per trial—e.g., too few CT angles for target PSNR (64 instead of 128) or insufficient k-space coverage for MRI. Mean PSNR drop: 2.1 dB vs. full pipeline.
- **Removing $\mathcal{A}_P$** (60 runs): human experts write `spec.md` manually, averaging 52 min per specification with format or physics errors in $\sim 14\%$ of cases. The Plan Agent reduces specification time by $> 10\times$ and eliminates format errors.
- **Template only**: fixed templates cover only 3/12 modalities (CT, MRI, lensless); all others fail outright.

The full pipeline is the only configuration achieving bounded-error success on all 12 modalities across all 5 trials. No single agent is dispensable.

2.10 Expert Comparison

We compared the agent framework against expert manual design on 6 modalities, 3 with real measured data (Table 6). Expert setup times are measured from specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation. For CT and MRI, expert times are cross-validated against published development reports [4, 10]; for remaining modalities, times are estimated from published timelines and confirmed by domain

258 experts.

Table 6. Agent framework vs. expert manual design. “Quality”: agent PSNR / expert PSNR. Bold = real measured data.

Modality	PSNR (dB)		Setup Time		Quality	LoC
	Agent	Expert	Agent	Expert		
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI (real)	31.7	32.4	20 min	5 d	97.8%	14 vs 520
CASSI (real)	29.5	30.2	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

259 Across all 6 modalities, the agent achieves $97.6 \pm 1.1\%$ of expert quality with ρ_{time} ranging from
 260 144 (DOT) to 420 (CT). Expert code ranges from 210 to 680 lines; agent specifications are
 261 10–15 lines. The quality gap is largest for DOT (95.3%), the most complex scattering chain,
 262 where operation at Tier-2 rather than Tier-3 contributes the largest $\varepsilon_{\text{unmod}}$; this gap closes under
 263 tier-lifting (Table 8).

264 2.11 Tier-Lifting: Cross-Tier Hybrid Reconstruction

265 Each of the 11 FPB primitives is implemented at all four fidelity tiers (Tier-0 through Tier-3),
 266 with dedicated adapter classes providing tier-specific forward models, adjoints, and validation
 267 checks (Table 7). The framework automatically selects the appropriate tier via a **TierPolicy**
 268 based on compute budget and modality requirements—for example, CT defaults to Tier-1/2,
 269 while NeRF requires Tier-3.

Table 7. Four-tier fidelity hierarchy. Each tier has a dedicated adapter with tier-specific validation checks. All 11 primitives are implemented at every tier; what changes is the physics model underlying each primitive.

Tier	Physics	Validation	Example
0 (geometric)	Ray optics, transforms	Invertibility, adjoint	LiDAR
1 (Fourier)	Paraxial, angular spectrum	Parseval energy, adjoint	SIM, holography
2 (shift-variant)	Radon, k-space, Mie	Positivity, reciprocity, cost	CT, MRI
3 (full-physics)	FDTD, BPM, MC, NeRF	Agreement, uncertainty calib.	DOT, NeRF

270 When a modality benefits from combining fidelity levels—e.g., using a fast Tier-1 adjoint for
 271 gradient steps but a Tier-3 forward model for accurate residual evaluation—the *Tier-Lifting*
 272 *Protocol* provides hybrid reconstruction via a Hybrid Proximal-Gradient (HPG) method:

273 **Proposition 3** (Tier-Lifting). *Let A_{low} be the low-tier forward model and A_{high} the high-tier*
 274 *model (same 11 primitives, different fidelity). Define $\Gamma(x) = A_{\text{high}}(x) - A_{\text{low}}(x)$. Then:*

$$\varepsilon_{\text{unmod}} \leq \left\| \Gamma(x) - \hat{\Gamma}(x) \right\| = \varepsilon_{\text{lift}}$$

275 where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{\text{lift}} = 0$ and
 276 the five-term error bound reduces to four terms.

277 The HPG loop:

$$x_{k+1} = \text{prox}_{\lambda R} \left(x_k - \gamma \cdot A_{\text{low}}^T (A_{\text{high}}(x_k) - y) \right)$$

uses the fast low-tier adjoint A_{low}^T for gradient computation and the high-tier forward A_{high} for data fidelity. Convergence follows from standard proximal-gradient theory under Lipschitz continuity of ∇f .

We validated the tier-lifting protocol on two cross-tier combinations (Table 8).

Table 8. Tier-lifting validation: cross-tier hybrid reconstruction eliminates model mismatch for both scattering and wave-optics domains.

Domain	Low tier	High tier	Correction	ϵ_{unmod}	ϵ_{lift}
DOT	Tier-2 (P_1 diffusion)	Tier-3 (P_3 RT)	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Tier-1 (angular spectrum)	Tier-3 (FDTD)	Evanescent + scatter	147%	$< 10^{-12}$

For DOT, the P_1 diffusion approximation (Tier-2) neglects higher-order angular moments; the P_3 correction (Tier-3) captures these via a coupled system, reducing ϵ_{unmod} to machine precision. For coherent imaging, the angular spectrum (Tier-1) misses evanescent waves and multiple scattering; FDTD (Tier-3) resolves all wave phenomena.

The tier-lifting protocol preserves the FPB structure: the same 11 primitives, the same DAG topology, the same `spec.md`—only the fidelity level of each primitive changes. The framework therefore natively spans all four tiers, from geometric optics to full-wave simulation and learned surrogates.

3Discussion

Limitations and caveats. Before interpreting the results, we note four limitations. (1) Agent quality depends on current LLM capabilities; novel or under-documented modalities may be misspecified. (2) Of 168+ modalities that compile, quantitative reconstruction is validated for 36; real-data validation covers 3 modalities (CT, MRI, CASSI). The remaining 120+ modalities pass compiler gates but lack per-modality reconstruction benchmarks—compilation alone does not guarantee reconstruction quality. (3) Expert comparison relies partly on estimated setup times from published timelines [4, 10]; prospective head-to-head studies would strengthen these claims. (4) The Performance Agent’s accuracy degrades for scattering-dominated systems (MAE 3.8 dB vs. 0.9 dB for well-characterized modalities).

The specification as exchange standard. Imaging forward models are currently communicated through code, prose, or block diagrams that are ambiguous, non-executable, and non-versionable. The `spec.md` format addresses this by decoupling the *what* (physical problem) from the *how* (numerical method) in a structured 7-field document. This enables non-expert scientists to design imaging systems via natural language, and makes specifications shareable, versionable, and peer-reviewable independently of code—a journal could require a `spec.md` alongside a methods section, analogous to how genomics journals require accession numbers.

Formal verification versus ad hoc design. The Judge differentiates this framework from both expert manual design (no formal validation) and black-box AI design (no physical constraints). A neural design tool might produce a plausible-looking system with no certificate; the Judge catches errors deterministically through the 6-gate compiler, including CFL stability, coercivity, and Lipschitz checks. Its 96% rejection precision on 150 test specifications indicates robust detection of structural, parametric, and noise-model errors, though subtle parameter range violations (1–5% outside bounds) remain a gap.

Bridging the reality gap. Theorem 2 converts the implicit, unmeasured gap between a designed model and reality into an explicit five-term error budget. Because each term is independently measurable, the theorem provides a diagnostic protocol: the practitioner can identify which term

dominates and take corrective action (calibration, specification refinement, or tier selection). In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; $\varepsilon_{\text{trans}} = 0$ for 86% of validated modalities; $\varepsilon_{\text{unmod}} = 0$ when operating at Tier-3 (full-physics) fidelity, and is reducible to machine precision via tier-lifting for lower tiers. The dominant residual term is therefore $\varepsilon_{\text{param}}$ (parameter mismatch), which is reducible by calibration.

Relation to existing work. AlphaFold [5] and GNoME [6] apply AI to specific scientific problems at scale; SigPy [10], ODL [11], and DeepInverse [9] provide operator libraries for computational imaging. Our framework provides the *design* layer that precedes operator instantiation, with formal error guarantees absent from both data-driven discovery and existing operator libraries.

Outlook. Extending real-data validation to additional modalities (ultrasound, OCT, PET), expanding the tier-lifting validation to more cross-tier combinations, improving Performance Agent calibration for scattering systems, and establishing a community benchmark for imaging system design are the natural next steps.

4Methods

4.1Proof of Theorem 2

Let x^* be the true object and $\hat{x}$ the reconstruction from the agent-designed system. Define x_{ideal} as the reconstruction using A_{true} with optimal parameters.

Step 1 (Model approximation). $\|A_{\text{agent}} - A_{\text{true}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}$ by the chain of approximations: true physics $\rightarrow$ selected-tier model ($\varepsilon_{\text{unmod}}$) $\rightarrow$ FPB decomposition (ε_{FPB}) $\rightarrow$ specification ($\varepsilon_{\text{spec}}$) $\rightarrow$ translated DAG ($\varepsilon_{\text{trans}}$) $\rightarrow$ parameterized operator ($\varepsilon_{\text{param}}$).

Step 2 (Reconstruction stability). For a regularized reconstruction $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x)$, the perturbation bound gives:

$$\|\hat{x} - x_{\text{ideal}}\| \leq \frac{1}{\alpha + \lambda} \|(A_{\text{agent}} - A_{\text{true}})(x^*)\|$$

where α is the coercivity constant of $A^T A$ and λ is the regularization strength.

Step 3 (Composition). $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{ideal}}\| + \|x_{\text{ideal}} - x^*\|$. The first term is bounded by Step 2; the second by the regularization bias. The five ε terms propagate multiplicatively through the Lipschitz constants of each stage.

Step 4 (Bounding each term). $\varepsilon_{\text{FPB}} < 0.01$ by the FPB theorem [2]. $\varepsilon_{\text{spec}} = 0$ when all 6 compiler gates pass. $\varepsilon_{\text{trans}} = 0$ when canonical chain matches the registry. $\varepsilon_{\text{param}} \leq L_A \cdot \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ where L_A is computed by the compiler’s Lipschitz analysis of nonlinear primitives. $\varepsilon_{\text{unmod}} = 0$ at Tier-3; for lower tiers, estimated empirically per modality from cross-tier comparison (see Table 8).

4.26-Gate Compiler Implementation

Gate 1 (DAG). The agent’s element list is compiled into an `OperatorGraphSpec` (typed Pydantic v2 model). DAG acyclicity is verified via Kahn’s algorithm in $O(V + E)$.

Gate 2 (Chain). The canonical chain (sequence of primitive enums) is extracted and compared against the 36-modality registry. Match is exact or within known equivalences (e.g., Σ merged into D).

Gate 3 (Bounds). Node count $N \leq 20$, DAG depth $D \leq 10$.

Gate 4 (Nonlinear). For each nonlinear node (D, R, Λ): family enum resolved, parameter count ≤ 2 , values within physics-based bounds, Lipschitz constant computed.

Gate 5 (Adjoint). Randomized dot-product test: $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) < 10^{-4}$, 3 trials.

Gate 6 (Error). When reference A_{true} is available: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| / \|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors, threshold < 0.01 .

4.3 Recovery Ratio Protocol

Four scenarios measure model mismatch impact:

$$\text{I (oracle): } y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (2)$$

$$\text{II (mismatch): } y = A_{\text{true}}(x) + n; \quad \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (3)$$

$$\text{III (auto-correct): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (4)$$

$$\text{IV (calibrated): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{III}} - \text{PSNR}_{\text{II}}) / (\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures auto-correction effectiveness.

4.4 Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101 implementations covering all 11 canonical types across all 4 fidelity tiers, each with a dedicated adapter class: **Tier0Adapter** (geometric: invertibility checks, ray transforms), **Tier1Adapter** (Fourier: Parseval energy conservation, Fresnel propagation, OTF convolution), **Tier2Adapter** (shift-variant: positivity, reciprocity, Mie scattering, cost estimation), and **Tier3Adapter** (full-physics: FDTD, Monte Carlo, learned surrogates with uncertainty calibration and agreement validation). A **TierPolicy** automatically selects the appropriate tier based on compute budget and modality-specific minimum requirements. Agents are LLM pipelines with tool access to the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is implemented as a generic **TierLiftingProtocol** class accepting any pair of low-tier/high-tier forward models, with validated cross-tier combinations for DOT (P_1/P_3) and coherent imaging (angular spectrum/FDTD). The 36-modality registry, compiler, 4-tier adapter library, tier-lifting module, and test suite are publicly available. Source code: https://github.com/integritynoble/Physics_World_Model.

Data Availability

The LoDoPaB-CT dataset is publicly available [3]. The M4Raw MRI dataset and KAIST TSA CASSI dataset are publicly available from their respective repositories. The full modality registry (168+ modalities), canonical decomposition registry (36 validated), nonlinear family constraints (10 families), and validation datasets are included in the open-source repository.

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, and the 101-primitive library—is available at <https://github.com/integritynoble/>

390 `Physics_World_Model` under the MIT licence.

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